

I. Linear Algebra

Core Properties & Subspaces

- Linear Independence:** $v_1, \dots, v_k$ indep. iff $c_1v_1 + \dots + c_kv_k = 0 \Rightarrow c_1 = \dots = c_k = 0$ (no nontrivial combo = 0).
- Invertibility Equivalence:** A ($n \times n$) invertible $\iff \det A \neq 0 \iff Ax = 0$ has only trivial sol. $\iff Ax = b$ unique $\forall b \iff$ cols indep. $\iff$ cols span $\mathbb{R}^n \iff$ full rank $\iff 0$ not an eigenvalue.
- Determinant (3×3):** $\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$.
- Det. Properties:** $\det(AB) = \det A \det B$; $\det A^T = \det A$; $\det A^{-1} = 1/\det A$; $\det(cA) = c^n \det A$. Row swap flips sign; scaling a row by c scales $\det$ by c ; row addition: no change. Triangular: $\det = \prod \text{diag}$.
- Cramer's Rule / Inverse:** $x_i = \det(A_i)/\det(A)$, $A_i = A$ w/ col $i \rightarrow b$. 2×2 : $A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- Transpose/Inverse of Product:** $(AB)^T = B^T A^T$; $(AB)^{-1} = B^{-1} A^{-1}$ (order reverses in both).
- Inverse (3×3):** $A^{-1} = \frac{1}{\det A} \text{adj}(A)$ ($\text{adj} = C^T$, cofactors). For $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$:
$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} ei - fh & ch - bi & bf - ce \\ fg - di & ai - cg & cd - af \\ dh - eg & bg - ah & ae - bd \end{pmatrix}$$
- Solution Parameterization:** $Ax = b$ consistent $\Rightarrow x = x_p + x_h$ (x_p particular, $x_h \in \text{Null}(A)$, free vars).
- Rank-Nullity:** $\dim(V) = \dim(\text{Im } T) + \dim(\text{Ker } T)$, $T: V \rightarrow W$.
- Subspace Bases:** Col space basis = pivot columns of A (original, not RREF). Null space basis: solve $Ax = 0$, parameterize free vars.
- Surjective/Injective Maps:** $T: V \rightarrow W$ surjective iff $\text{Im } T = W$; injective iff $\text{Ker } T = \{0\}$. $\dim V < \infty$, $T: V \rightarrow V$: injective $\iff$ surjective $\iff$ invertible.
- Rank 1 Matrices:** $A \neq 0$ ($m \times n$) has rank 1 iff $A = cd^T$, $c \in \mathbb{R}^m$, $d \in \mathbb{R}^n$.
- Direct Sum:** $V = U \oplus W$ iff $V = U + W$ & $U \cap W = \{0\}$ (unique decomp. $v = u + w$); $\dim V = \dim U + \dim W$.

Eigendecomposition & Orthogonality

- Eigenvalues & Eigenvectors:** $Av = \lambda v$ ($v \neq 0$). λ : roots of $\det(A - \lambda I) = 0$. $\text{tr}(A) = \sum \lambda_i$, $\det(A) = \prod \lambda_i$. 2×2 shortcut: $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$.
- Alg. vs. Geo. Multiplicity:** Alg. mult. = mult. as root of char. poly; geo. mult. = $\dim \text{Ker}(A - \lambda I)$. $1 \leq \text{geo} \leq \text{alg}$. Diagonalizable iff $\text{geo} = \text{alg} \forall \lambda$ (else e.g. Jordan blocks).
- Diagonalization:** $A = XDX^{-1}$: X = indep. eigenvectors, D = eigenvalues (diag).
- Matrix Powers:** $A^k = XD^kX^{-1}$. All $|\lambda_i| < 1 \Rightarrow D^k \rightarrow 0 \Rightarrow A^k \rightarrow 0$.
- Spectral Theorem:** $A = A^T \Rightarrow A = Q\Lambda Q^T$ ($Q^T Q = I$). Eigenvalues real; eigenvectors of distinct λ 's orthogonal.
- Quadratic Forms & Definiteness:** $Q(x) = x^T Ax$ (A symmetric). PD: $> 0 \forall x \neq 0$; ND: < 0 ; PSD/NSD: $\geq 0 / \leq 0$; indefinite: both signs. (Tests: see Optimization.)
- Rayleigh Quotient:** A symmetric: $\max_{\|u\|=1} u^T Au = \lambda_{\max}$, $\min_{\|u\|=1} u^T Au = \lambda_{\min}$; more gen. $\lambda_{\min} \|x\|^2 \leq x^T Ax \leq \lambda_{\max} \|x\|^2$.
- Similarity Transformations:** S invertible $\Rightarrow T, S^{-1}TS$ same eigenvalues; $Tv = \lambda v \Rightarrow S^{-1}TS(S^{-1}v) = \lambda(S^{-1}v)$.
- Gram-Schmidt:** $e_1 = v_1/\|v_1\|$; $u_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j$, $e_k = u_k/\|u_k\|$.
- Idempotent Matrices:** $P^2 = P$. Eigenvalues $\in \{0, 1\}$ only (since $\lambda^2 = \lambda$). Projection $P = X(X^T X)^{-1} X^T$ ($X: n \times k$), $\text{tr}(P) = k$; residual $M = I_n - P$, $\text{tr}(M) = n - k$.
- Norms & Inner Products:** $\langle u, v \rangle = 0 \iff \|u\| \leq \|u + av\| \forall a$. Norms: $\|x\|_2 = (\sum x_i^2)^{1/2}$, $\|x\|_\infty = \max |x_i|$, $\|x\|_1 = \sum |x_i|$.
- Involutions:** $T^2 = I$ (T = own inverse). Eigenvalues: $\lambda = \pm 1$ only.
- Nilpotency:** $A^k = 0$ some k . All eigenvalues = 0; never invertible, never diagonalizable (unless $A = 0$). Neumann: $(I - N)^{-1} = I + N + \dots + N^{k-1}$; $(I + N)^{-1} = I - N + \dots + (-1)^{k-1} N^{k-1}$.
- Invariant Subspace:** $W \subseteq V$ invariant under T iff $T(W) \subseteq W$. E.g. eigenspaces $\text{Ker}(A - \lambda I)$.

II. Real Analysis

Topology & Metric Spaces

Sets, Compactness & Boundedness

- Set Operations & De Morgan's Laws:** Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$. Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. **De Morgan:** $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$; gen. (incl. infinite): $(\bigcup_i A_i)^c = \bigcap_i A_i^c$, $(\bigcap_i A_i)^c = \bigcup_i A_i^c$.
- Sup/Inf & Completeness:** $\sup S$ = least upper bd., $\inf S$ = greatest lower bd. Completeness: $S \subseteq \mathbb{R}$ bdd above $\Rightarrow \sup S$ exists (distinguishes $\mathbb{R}$ from $\mathbb{Q}$) $\Rightarrow$ Archimedean: $\forall x \exists n \in \mathbb{N}, n > x$.
- Bolzano-Weierstrass:** Bounded seq. in $\mathbb{R}^n \Rightarrow$ convergent subsequence exists (via completeness + nested-interval bisection).
- Open and Closed Sets:** U open: $\forall p \in U \exists r > 0, B_r(p) \subseteq U$. Closed: complement open (= contains all limit pts). Cont. preimages preserve open/closed.
- Closure of a Set:** $\bar{S} = S \cup \{\text{limit pts.}\}$ = smallest closed set $\supseteq S$. S closed iff $S = \bar{S}$.
- Open Balls:** $B_\epsilon(x) = \{y \in \mathbb{R}^n : \|y - x\| < \epsilon\}$. U open iff $\forall x \in U \exists \epsilon, B_\epsilon(x) \subseteq U$.
- Compactness (Heine-Borel):** In $\mathbb{R}^n$: compact $\iff$ closed+bounded $\iff$ every open cover has finite subcover $\iff$ every seq. has convergent subseq.
- Nested Compact Sets:** $K_1 \supseteq K_2 \supseteq \dots$ non-empty compact $\Rightarrow \bigcap K_k \neq \emptyset$.
- Bounded Set:** S bounded iff $\sup_{x,y \in S} d(x,y) < \infty$ iff $S \subseteq B_M(x_0)$ some x_0, M .
- Limit Point:** a limit pt. of S : every $B_r(a)$ contains infinitely many pts of S . Closed sets contain all their limit pts.
- Separation of Hyperplanes:** Disjoint convex $C_1, C_2 \subseteq \mathbb{R}^n$ (one compact, other closed) $\Rightarrow \exists p \neq 0, c$: $p^T x \leq c \leq p^T y \forall x \in C_1, y \in C_2$.

Convergence, Contraction & Fixed Points

- Convergence & Cauchy:** $x_n \rightarrow L$: $\forall \epsilon \exists N, \forall n \geq N, d(x_n, L) < \epsilon$. Cauchy: $d(x_n, x_m) < \epsilon$ for large n, m .
- Contraction Maps:** $d(Tx, Ty) \leq k d(x, y)$, $k \in [0, 1)$.
- Banach Fixed Point:** (X, d) complete, T contraction $\Rightarrow$ unique fixed pt $x^* = Tx^*$; $x_{n+1} = Tx_n \rightarrow x^*$ from any x_0 (e.g. Bellman eqns).

Continuity & Functions

- Continuous (ϵ - δ):** f cont. at c : $\forall \epsilon \exists \delta, d_X(x, c) < \delta \Rightarrow d_Y(f(x), f(c)) < \epsilon$.
- Sequential Continuity:** f cont. at c iff $\forall x_n \rightarrow c, f(x_n) \rightarrow f(c)$ (equiv. to ϵ - δ ; easiest way to disprove continuity).
- Uniform Continuity:** δ depends only on ϵ , not the point. Cont. on compact domain $\Rightarrow$ unif. cont. (not conversely; e.g. x^2 on $\mathbb{R}$).
- Topological Continuity:** f cont. iff $f^{-1}(U)$ open $\forall$ open U .
- IVT:** f cont. on $[a, b]$, y between $f(a), f(b) \Rightarrow \exists c \in [a, b], f(c) = y$.
- Extreme Value Theorem:** f cont., K non-empty compact $\Rightarrow f$ attains max & min on K (existence argument for optima).
- Well-Defined Function:** Each domain elt. maps to a unique codomain elt. Cont. transforms (e.g. $|\cdot|$) preserve well-definedness.
- Surjective/Injective Functions:** $f: X \rightarrow Y$: **injective** iff $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; **surjective** iff $f(X) = Y$; **bijective** iff both (inverse f^{-1} exists). Composition preserves injectivity/surjectivity.

Metric Spaces, Norms & Inner Products

- Metric Space:** (X, d) : $d: X \times X \rightarrow \mathbb{R}_+$ s.t. $d(x, y) = 0 \iff x = y$; symmetric; $d(x, z) \leq d(x, y) + d(y, z)$.
- Discrete Metric:** $\rho(x, y) = 1$ if $x \neq y$, else 0. All subsets clopen & bounded; infinite sets not compact; seqs. converge only if eventually constant.
- Norms, Inner Products, Distance:** Inner product: symmetric, bilinear, pos-def. Induces norm $\|x\| = \sqrt{\langle x, x \rangle}$; **Cauchy-Schwarz:** $|\langle x, y \rangle| \leq \|x\| \|y\| \Rightarrow$ triangle ineq. Norm induces metric $d(x, y) = \|x - y\|$ (not conversely).
- Norm Equivalence:** All norms on $\mathbb{R}^n$ equiv.: $\exists c, C > 0, c\|x\|_a \leq \|x\|_b \leq C\|x\|_a \forall x$ (e.g. $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\|x\|_\infty$). Conv./open/compact don't depend on norm choice.

Calculus & Differentiation

- Derivative Definition:** $f(x+h) = f(x) + Df(x)h + r(h)$, $\lim_{h \rightarrow 0} \|r(h)\|/\|h\| = 0$. Cont. partials in nbhd. $\Rightarrow$ diff'ble.
- Basic Derivative Formulae:** Product: $(fg)' = f'g + fg'$. Quotient: $(f/g)' = \frac{gf' - fg'}{g^2}$. Chain: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$. $\ln'(x) = 1/x$, $(e^x)' = e^x$. $\sin' = \cos$, $\cos' = -\sin$, $\tan' = \sec^2$.
- Taylor Approximation (Multivariate):** $f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2}h^\top H_f(x)h + o(\|h\|^2)$ as $h \rightarrow 0$. **Drop remainder:** $f(x+h) \approx f(x) + \nabla f(x)^\top h + \frac{1}{2}h^\top H_f(x)h$ — safe when $\|h\|$ small (local approx.; error vanishes faster than $\|h\|^2$), not for large h .
- Scalar Mean Value Theorem:** $f(b) - f(a) = Df(c)(b-a)$, c on segment $[a,b]$.
- L'Hôpital:** $\frac{0}{0}$ or $\frac{\infty}{\infty} \Rightarrow \lim f/g = \lim f'/g'$ (if latter exists).
- Convex Sets & Concave/Convex Functions:** C convex: $tx + (1-t)y \in C \forall t \in [0,1]$. f **concave**: $f(tx + (1-t)y) \geq tf(x) + (1-t)f(y)$ (convex: reverse). Twice-diff.: concave iff H_f NSD (convex iff PSD); strict definiteness $\Rightarrow$ strict (not $\Leftarrow$).
- Integrals: FTC:** $\int_a^b f dx = F(b) - F(a)$ ($F' = f$). **By parts:** $\int u dv = uv - \int v du$. **Sub:** $\int f(g(x))g'(x)dx = \int f(u)du$. **Leibniz:** $\frac{d}{dt} \int_{a(t)}^{b(t)} f(x,t)dx = f(b(t),t)b'(t) - f(a(t),t)a'(t) + \int_{a(t)}^{b(t)} f_t dx$. **Fubini:** $\iint_R f dA = \int_a^b \int_c^d f dy dx$. **Improper:** $\int_a^\infty f dx = \lim_{b \rightarrow \infty} \int_a^b f dx$.

Multivariable Differential Calculus

- Partial and Total Derivatives:** $Df(x)$: linear op. s.t. $f(x+h) = f(x) + Df(x)h + r(h)$, $\|r(h)\|/\|h\| \rightarrow 0$. Cont. partials in nbhd. $\Rightarrow$ diff'ble; $Df(x)$ = Jacobian.
- Multivariate Derivative Rules:** Chain rule: $D(h \circ g \circ f) = Dh \cdot Dg \cdot Df$. Directional deriv.: $D_v f(x) = \nabla f(x)^\top v$. $\nabla(fg) = g\nabla f + f\nabla g$. $\nabla(x^\top Ax) = (A + A^\top)x$ ($= 2Ax$ if symmetric). ∇f level set $\{f = c\}$; points toward steepest increase.
- Multivariate Mean Value Theorem:** Scalar f : $f(b) - f(a) = \nabla f(c)^\top (b-a)$, $c \in [a,b]$. Vector-valued ($m > 1$): no exact c ; only $\|f(b) - f(a)\| \leq \sup_{[a,b]} \|Df\| \|b-a\|$.
- Implicit Function Theorem:** $F(x,y) = 0$, $F_y \neq 0 \Rightarrow$ locally $y = g(x)$, $g_x = -F_x/F_y$, $F(x,y,z) = 0$, $F_z \neq 0 \Rightarrow z = g(x,y)$, $g_x = -F_x/F_z$, $g_y = -F_y/F_z$.
- Implicit Fn. Thm. — Steps:** (1) Check $F(x_0,y_0) = 0$. (2) Compute $D_y F$ (Jacobian in endog. vars y), verify $\det D_y F(x_0,y_0) \neq 0$ (scalar: $F_y \neq 0$). (3) Conclude: unique $C^1 y = g(x)$ near x_0 , $g(x_0) = y_0$, $F(x,g(x)) \equiv 0$. (4) Chain rule on identity: $D_x F + D_y F Dg = 0 \Rightarrow Dg = -[D_y F]^{-1} D_x F$ (scalar: $g_x = -F_x/F_y$). (5) Plug in (x_0,y_0) for numeric slope; differentiate again for 2nd-order/Taylor terms.
- Inverse Function Theorem:** $F \in C^1$, $\det DF(x) \neq 0 \Rightarrow$ local diffeomorphism (diff'ble local inverse). Sufficient for local, not global, invertibility.
- Inverse Fn. Thm. — Steps:** (1) Confirm $f \in C^1$ near x_0 . (2) Compute $Df(x_0)$, verify $\det Df(x_0) \neq 0$. (3) Conclude: nbhds $V \ni x_0, W \ni f(x_0)$ with $f: V \rightarrow W$ a C^1 -diffeomorphism (bijective, C^1 inverse). (4) $D(f^{-1})(y) = [Df(x)]^{-1}$, $x = f^{-1}(y)$ — invert the Jacobian (e.g. 2×2 formula). (5) Local only: $\det DF \neq 0$ everywhere $\Rightarrow$ global injectivity — check separately (e.g. periodicity).

III. Optimization & Economic Applications

- First and Second Order Conditions:** Local optimum: $\nabla f(x) = 0$ (necessary). H_f PD $\Rightarrow$ strict min; ND $\Rightarrow$ strict max; indefinite $\Rightarrow$ saddle.
- Definiteness Testing:** A symmetric. **Eigenvalue test:** PD iff all $\lambda > 0$; ND iff all < 0 ; PSD/NSD iff all $\geq 0 / \leq 0$; indefinite iff mixed signs. **Leading minors** D_k : PD iff $D_k > 0 \forall k$; ND iff $(-1)^k D_k > 0 \forall k$. (Semidefiniteness needs **all** principal minors, not just leading.)
- Lagrangian Formulation:** $g(x) \geq 0, h(x) = 0$: $\mathcal{L} = f(x) - \lambda^\top h(x) + \mu^\top g(x)$. λ (equality) free in sign; μ (inequality) ≥ 0 .
- KKT Conditions:** At optimum x^* (w/ CQ): (i) **Stationarity** $\nabla_x \mathcal{L} = 0$; (ii) **Primal feas.** $g \geq 0, h = 0$; (iii) **Dual feas.** $\mu \geq 0$; (iv) **Compl. slackness** $\mu_i g_i(x^*) = 0 \forall i$.

- Solving a Lagrangian — Steps:** (1) $\mathcal{L} = f(x) - \lambda^\top h(x) + \mu^\top g(x)$. (2) FOC: $\nabla_x \mathcal{L} = 0$. (3) Guess slack ($\mu = 0$) for constraints expected non-binding; solve, check $g(x) \geq 0$. (4) If infeasible, set binding $g_i(x) = 0$, re-solve treating μ_i unknown. (5) Verify compl. slack. $\mu_i g_i = 0$, dual feas. $\mu \geq 0$, primal feas. $g \geq 0, h = 0$. (6) Confirm optimum: bordered Hessian / SOC, or concavity+Slater $\Rightarrow$ KKT suff.
- Constraint Qualifications: LICQ:** binding constraint grads. lin. indep. $\Rightarrow$ KKT necessary. **Slater** (convex probs): $\exists \hat{x}, g_j(\hat{x}) > 0 \forall j, h_i(\hat{x}) = 0$ (weaker).
- Bordered Hessian (Constrained SOC):** n vars, m eq. constraints: $\bar{H} = \begin{pmatrix} 0 & Dh^\top \\ Dh & H_{\mathcal{L}} \end{pmatrix}$. Last $n-m$ leading minors (orders $2m+1..n+m$): **max** alternate from $(-1)^{m+1}$; **min** all sign $(-1)^m$.
- Envelope Theorem:** $\frac{dV}{dm} = \frac{\partial \mathcal{L}}{\partial m}$ (value fn.'s marginal change = Lagrangian's partial).
- Homogeneous & Homothetic Functions:** $f(tx) = t^k f(x)$: homog. degree k . Homothetic = monotonic transform of homog. fn.
- Quasi-concavity:** $\{f \geq \alpha\}$ convex $\forall \alpha$ (iff $f(tx + (1-t)y) \geq \min\{f(x), f(y)\}$). Weaker than concave (concave $\Rightarrow$ QC, not conversely; e.g. utility fns.).
- Additivity & Divisibility:** $f(x+y) \geq f(x) + f(y)$ & $f(tx) \geq tf(x) \forall t \in [0,1] \Rightarrow f$ concave & homog. degree 1.
- Log-Linearization:** $\hat{z} = (z - z^*)/z^*$ (% deviation from steady state).
- Correspondences (Set-Valued Maps):** Key properties: upper/lower hemicontinuity; closed/compact/convex-valued. Compact graph $\Rightarrow$ compact-valued (not conversely).

IV. Probability

- Kolmogorov's Probability Axioms:** Ω arbitrary sample space; $\mathcal{F} \subseteq 2^\Omega$: σ -algebra (closed under complement, countable union); $P: \mathcal{F} \rightarrow [0,1]$ countably additive, $P(\Omega) = 1$.
- Derived / Required Properties:** **Null:** $P(\emptyset) = 0$; **Norm.:** $P(\Omega) = 1$; **Complement:** $P(A^c) = 1 - P(A)$; **Countable additivity:** disjoint $\{A_i\}$: $P(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty P(A_i)$; **Translation inv.:** $P(x+A) = P(A)$; **Union:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; **Monotone:** $A \subseteq B \Rightarrow P(A) \leq P(B)$.
- CDF and PMF/PDF:** Discrete: PMF $p(x) = P(X=x)$. Continuous: PDF $f \geq 0$, $P(a \leq X \leq b) = \int_a^b f dx$. CDF $F(x) = P(X \leq x)$: non-decr., right-cont., $F(-\infty) = 0, F(\infty) = 1, F' = f$.
- Conditional Probability:** $P(A | B) = \frac{P(A \cap B)}{P(B)}$, $P(B) > 0$.
- Bayes' Rule:** $P(A | B) = \frac{P(B | A)P(A)}{P(B)}$.
- Law of Total Probability:** $\{B_i\}$ partition of Ω : $P(A) = \sum_i P(A | B_i)P(B_i)$.
- Independence:** A, B indep. iff $P(A \cap B) = P(A)P(B)$ iff $P(A | B) = P(A)$.
- Independence of Random Variables:** X, Y indep. iff joint = product of marginals ($F_{X,Y} = F_X F_Y$) $\Rightarrow E[XY] = E[X]E[Y]$, $\text{Cov}(X,Y) = 0$ (converse false).
- Expectation:** $E[X] = \sum xp(x)$ (discrete) or $\int xf(x)dx$ (cont.); linear: $E[aX + bY] = aE[X] + bE[Y]$.
- Conditional Expectation:** $E[X | Y = y] = \sum xp(x | y)$.
- LIE:** $E[X] = E[E[X | Y]]$.
- Variance and Covariance:** $\text{Var}(X) = E[X^2] - E[X]^2$. $\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$. $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X,Y)$.
- Jensen's Inequality:** g convex $\Rightarrow E[g(X)] \geq g(E[X])$ (concave: reverse). E.g. $E[X^2] \geq (E[X])^2$.
- Sample Variance:** $S_n^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$ unbiased: $E[S_n^2] = \sigma^2$ ($n-1$ corrects for est. μ by $\bar{X}_n$).
- Markov's Inequality:** $X \geq 0, a > 0$: $P(X \geq a) \leq E[X]/a$.
- Chebyshev's Inequality:** $P(|X - E[X]| \geq k) \leq \text{Var}(X)/k^2$.
- Law of Large Numbers:** i.i.d. X_i , mean μ : $\bar{X}_n \rightarrow \mu$ (WLLN: in prob.; SLLN: a.s.).
- Central Limit Theorem:** i.i.d. X_i , mean μ , var σ^2 : $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$.